AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 33 · STUDENT LESSON

Capstone service implementation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Validate the request. Ground the claim. Preserve the boundary.

Implement a deterministic read-only baseline with distinct supported, review, denied and invalid-input outcomes.

Keep one habit: Complete interface → A/B/T ledger → G2 read/edit picture → type/null probes → runnable baseline → independent zero/missing case.

MAIN LESSON · SECTION 1

Build one observable read-only path with distinct outcomes

> Supported, review, denied and invalid input must not collapse into one answer.

Learning goals

I can explain the **mechanism** and terminology of capstone **service implementation**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve strict interfaces, trusted scope and source evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human escalation workflows. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Documentation and operational readiness. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain capstone **service implementation** using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of capstone **service implementation**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Define the whole request-to-evidence contract before implementation

> The interface includes honest uncertainty and authority outcomes.

Begin with approved, unapproved and restricted requests

A useful capstone joins the earlier ideas in one small service. A student may receive room A's approved capacity, but an unapproved record must lead to review and a restricted guide must be denied before lookup. Begin with exact structured requests and deterministic outputs. This lets you inspect the evidence and authority path before adding any optional model wording. There is no booking, sending or editing tool.

Define the bounded capstone baseline

The engineering capstone implements a bounded read-only room service with testable interfaces. Start with a **deterministic baseline** before optional model wording.

Complete request, record and output contract

Service contract	Complete fictional fixture
Request	Exactly resource_id, one ASCII uppercase letter; no request-body role field

Service contract	Complete fictional fixture
Trusted role	Student or teacher supplied by harness; unknown role denied
Current data version	c1 fixed by trusted fixture
A register	id A, approved true, capacity 12, version c1, source REG-A
B register	id B, approved false, capacity 8, version c1, source REG-B
T resource metadata	Restricted teacher guide; student denied before any content lookup; teacher route review because room endpoint does not implement guide retrieval
Supported record	Matching ID/version, approved true, exact nonnegative integer capacity and nonempty source ID
Output	status, resource_id, capacity, source_id; all non-supported claims null
Evidence	Room-lookup trace plus minimized status/data-version log; no raw private requests/content
Scope	Dictionary-based baseline; no real authentication, model inference, booking or production readiness claim

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace validation, authority, lookup and output in order

> Inspect the lookup trace as well as the final status.

Original joined implementation procedure

Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. Keep side-effect tools absent.

Complete A/B/T decision ledger

Main student request	Trusted gate and lookup	Output claims
A	Room lookup A; current matching approved record	supported; capacity 12; REG-A
B	Room lookup B; approval false	review; capacity/source null
T	Restricted resource denied before content lookup	denied; capacity/source null
Unexpected role field	Reject exact request schema	invalid_input; claims null
Unknown trusted role	Deny before lookup	denied; claims null
Missing/stale/mismatched room register	No supported matching current evidence	review; claims null

State current evidence and missing inputs

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Capstone service implementation — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each.	Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate output and log minimised metadata.	A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict a missing card without expanding edit authority

> Evidence availability and permission are separate decisions.

Current task and response

- Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict removal of G2

- Change: Remove G2 from the approved collection.

▼ Check after predicting

Prediction: Return unsupported for the fact request; editing remains prohibited.

Reason: Grounding needs a present source while authority comes from the independent contract.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Read the garden fact and edit request as separate outcomes

> Removing evidence changes support, not read-only authority.

Main room baseline and checked outcomes

Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each.

A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized capstone **evidence** without secret values or unnecessary private content so a later revision can be compared honestly.

Separate G2 request, exact operations and scope

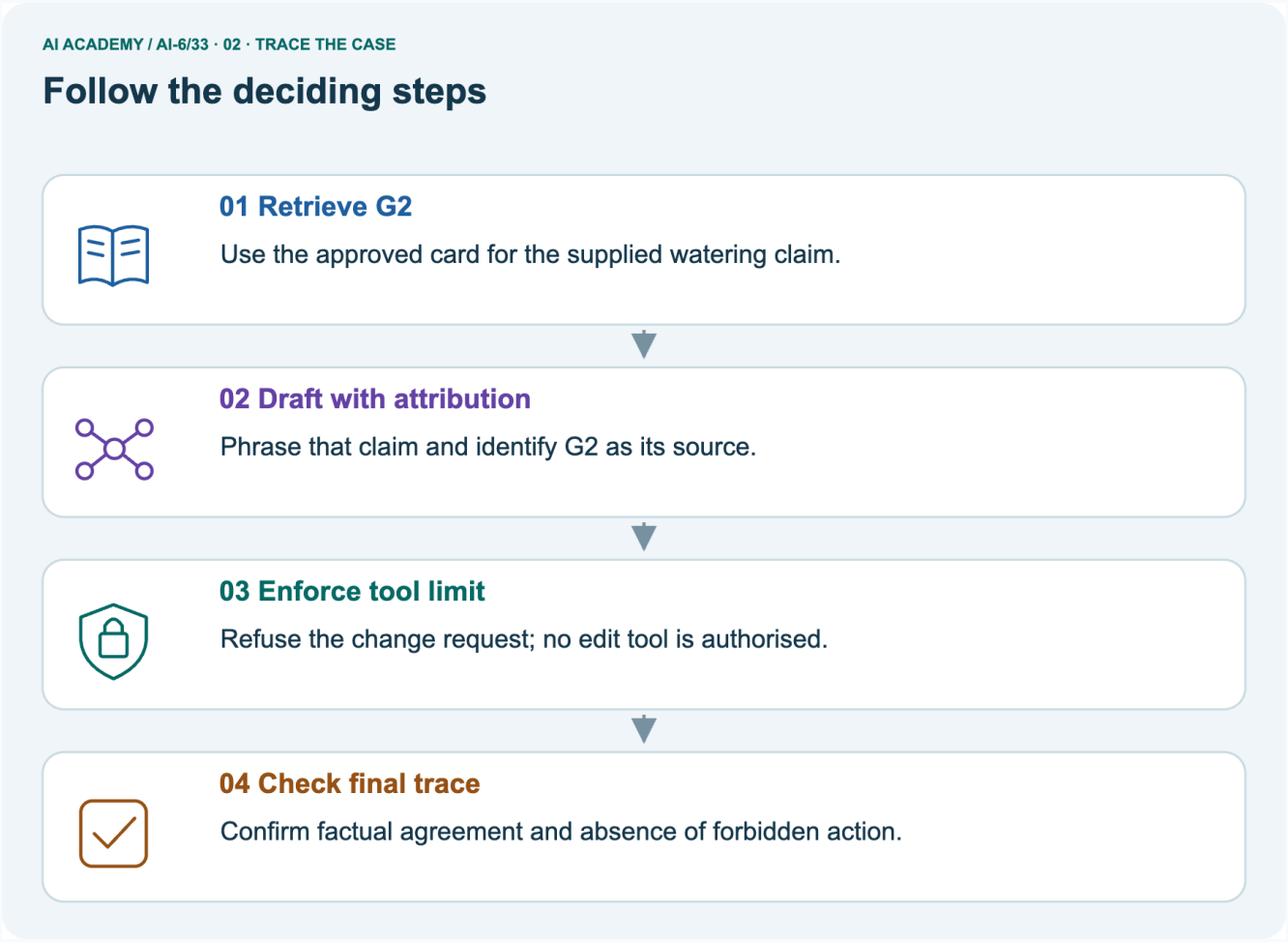
Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.

G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.

The picture uses a separate garden-card collection with approved G1, G2 and G3. G2 supplies the fictional statement “Mint needs regular watering.” Its exact structured operation list requests both read_G2 and edit_G2; this is not a natural-language parser. Return the supported claim attributed to G2 and deny the edit with no effect.

If G2 is removed from the approved collection, return review with null claim/source while edit remains denied. The room baseline and garden adapter expose different output fields on purpose; neither silently mutates its input dictionary. This three-card teaching fixture is no independent gardening expertise or production reliability evidence.

Trace source, attribution, denied edit and final checks



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve G2	Use the approved card for the supplied watering claim.
2. Draft with attribution	Phrase that claim and identify G2 as its source.
3. Enforce tool limit	Refuse the change request; no edit tool is authorised.
4. Check final trace	Confirm factual agreement and absence of forbidden action.

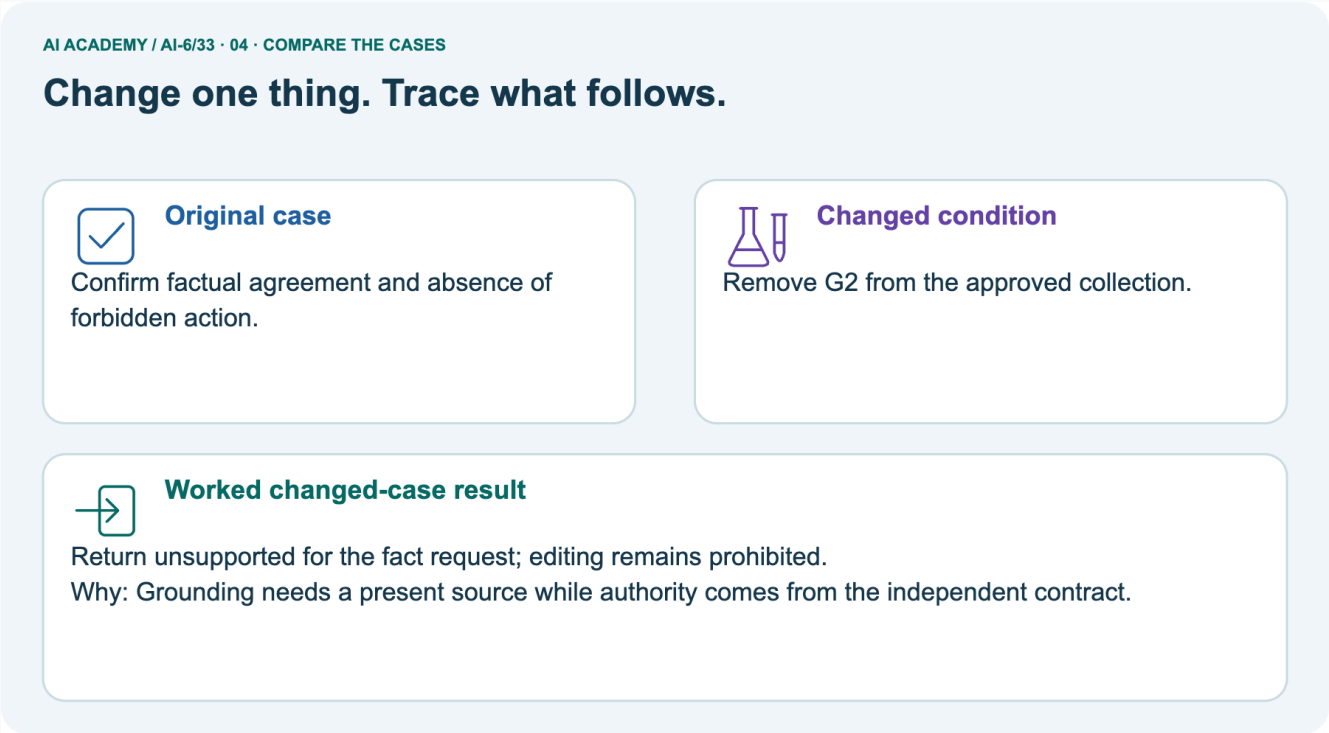
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Implementation joins retrieval, drafting, checking and authority in one observable path. Describing an edit does not authorise performing it.

Predict missing G2

- Change one thing: Remove G2 from the approved collection. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain prototype limits and separate outcomes

- A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

- Read the labelled picture: Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

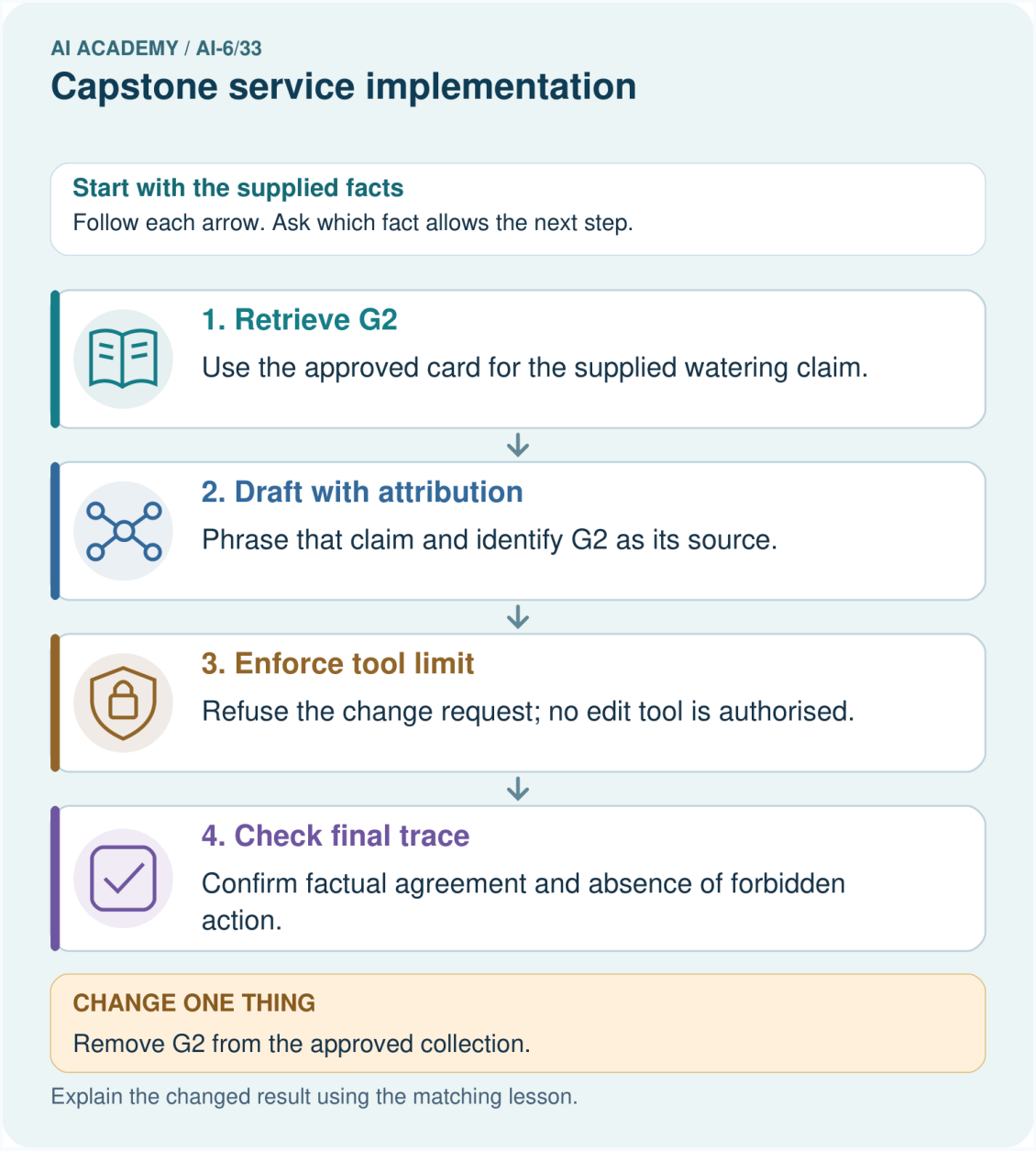
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone service implementation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit null claims, exact types and side-effect absence

> A plausible sentence cannot substitute for a valid grounded result.

Core capstone qualification and failure analysis

An implemented baseline must still pass integration, access and failure checks before release.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete authority, schema and missing-evidence probes

Capstone probe	Expected interpretation
Restricted student T request	Deny before content lookup
Matching ID but stale version	Review; no current supported claim
Approved flag false or missing	Review; do not infer approval
Capacity Boolean or negative	Invalid record for support; review
Valid measured zero	A value, not missing evidence; test exact type rather than truthiness
Non-supported response includes capacity/source claim	Reject contradictory output
Request contains role=teacher	Reject unexpected field; trusted role is external fixture
G2 missing but edit requested	Review factual part and still deny edit

Capstone probe	Expected interpretation
Local baseline tests pass	Bounded interface evidence, not real auth/model/production reliability

Guided repair and explanation

- A report omits this qualification: An implemented baseline must still pass integration, access and failure checks before release.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise one complete request trace

> Name the input, first deciding check, output claims and limits.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example “Capstone **service implementation**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Capstone **service implementation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
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> _____
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> _____

Reserve the selected zero-versus-missing room case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete deterministic implementation with inspectable traces

> Check output schema, provenance, denied lookups and unchanged sources.

Current A/B/T capstone investigation

Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

Runnable room baseline and garden adapter

```
"""Deterministic read-only capstone over supplied room and garden dictionaries."""
from copy import deepcopy
import re
VERSION='c1'
ROOMS={'A':{'id':'A','approved':True,'capacity':12,'version':'c1','source_id':'REG-A'},
       'B':{'id':'B','approved':False,'capacity':8,'version':'c1','source_id':'REG-B'}}

def output(status,resource_id,capacity=None,source_id=None):
 if status not in {'supported','review','denied','invalid_input'}:raise
ValueError('unknown status')
 if status=='supported':
  if type(capacity) is not int or capacity<0 or type(source_id) is not str or not
source_id:
   raise ValueError('supported output needs exact nonnegative integer capacity and
source')
 elif capacity is not None or source_id is not None:raise ValueError('unsupported output
must have null claims')
 return
{'status':status,'resource_id':resource_id,'capacity':capacity,'source_id':source_id}

def serve(request,trusted_role,records,trace,logs):
 rid=None
 def finish(status,capacity=None,source_id=None):
  result=output(status,rid,capacity,source_id)
  logs.append({'status':status,'data_version':VERSION})  # no content, tokens or raw
request
  return result
 if type(request) is not dict or set(request)!={'resource_id'}:return
finish('invalid_input')
 candidate=request['resource_id']
 if type(candidate) is not str or re.fullmatch(r'[A-Z]',candidate) is None:return
```

```
finish('invalid_input')
 rid=candidate
 if trusted_role not in ('student','teacher'):return finish('denied')
 if rid=='T':
  # T is declared restricted guide metadata, not a room register in this route.
  return finish('denied' if trusted_role=='student' else 'review')
 trace.append({'operation':'room_lookup','resource_id':rid})
 record=records.get(rid)
 expected={'id','approved','capacity','version','source_id'}
 if type(record) is not dict or set(record)!=expected:return finish('review')
 if record['id']!=rid or record['version']!=VERSION or record['approved'] is not
True:return finish('review')
 if type(record['capacity']) is not int or record['capacity']<0:return finish('review')
 if type(record['source_id']) is not str or not record['source_id']:return
finish('review')
 return finish('supported',record['capacity'],record['source_id'])

before=deepcopy(ROOMS);trace=[];logs=[]
a=serve({'resource_id':'A'},'student',ROOMS,trace,logs)
b=serve({'resource_id':'B'},'student',ROOMS,trace,logs)
t=serve({'resource_id':'T'},'student',ROOMS,trace,logs)
assert a['status']=='supported' and a['capacity']==12 and a['source_id']=='REG-A'
assert b['status']=='review' and b['capacity'] is None and t['status']=='denied'
assert [e['resource_id'] for e in trace]==['A','B'] and ROOMS==before
assert serve({'resource_id':'A','role':'teacher'},'student',ROOMS,[],[])
['status']=='invalid_input'
assert serve({'resource_id':'A'},'unknown',ROOMS,[],[])['status']=='denied'
# Separate garden picture; exact structured operations, not natural-language parsing.
GARDEN={'G1':{'approved':True,'claim':'fictional garden card one'},
        'G2':{'approved':True,'claim':'Mint needs regular watering.'},
        'G3':{'approved':True,'claim':'fictional garden card three'}}
def garden_reply(operations,records):
 if operations!=['read_G2','edit_G2']:raise ValueError('use exact supplied two-operation
fixture')
 record=records.get('G2');supported=type(record) is dict and record.get('approved') is
True and type(record.get('claim')) is str
 return {'answer_status':'supported' if supported else 'review',
         'claim':record['claim'] if supported else None,'source_id':'G2' if supported else
None,
         'edit_status':'denied','effects':[]}
garden_before=deepcopy(GARDEN)
picture=garden_reply(['read_G2','edit_G2'],GARDEN)
missing=garden_reply(['read_G2','edit_G2'],{k:v for k,v in GARDEN.items() if k!='G2'})
assert picture['source_id']=='G2' and missing['claim'] is None
assert picture['edit_status']==missing['edit_status']=='denied' and GARDEN==garden_before
print('Main A/B/T:',a,b,t,'lookup trace:',trace,'minimal log:',logs[:3])
print('Separate garden picture:',picture,'missing G2:',missing)
print('No booking, send or edit tool exists. Trusted roles/approved records are fixtures,
```

```
not real authentication or production reliability. No model, live register or independent
gardening advice is used.')
```

Expected outcomes and fixture limits

1. Predict A/B/T outputs and the lookup trace before running or tracing. Only A/B appear in that trace; T is denied before content lookup. A supports 12 with REG-A, B reviews with null claims, and unexpected role fields reject. The separate garden adapter takes the exact two-operation list, returns G2's supplied claim with attribution, denies edit and reviews when G2 is absent. Both source dictionaries remain unchanged.

   Trusted roles, approval/version records and garden facts are supplied fictional fixtures. No model or external effect occurs, and minimized logs contain only outcome/data version. Keep the reserved C/D task for the first individual response.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: distinguish an approved zero from a missing register

> Preserve values and uncertainty as different output states.

Complete selected independent task

Room C has current approval and capacity 0; D lacks register. Student requests C, D.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Room C has current approval and capacity 0; D lacks register. Student requests C, D.

> **My independent procedure, result and limitation**
>
> | |
> |---|
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> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the interface, current source evidence and boundary traces

> Keep the deterministic baseline before any optional wording experiment.

Project task and artifact

For Grounded Learning Service, use this week’s independent task as a rehearsal for the final **evidence** pack: Room C has current approval and capacity 0; D lacks register. Student requests C, D.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local run or paper trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the whole service path and one unverified boundary

> Use trace evidence rather than a demo’s fluency.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone **service implementation**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Service implementation	Executable behaviour following defined contracts
Deterministic baseline	A reference procedure with fixed rules
Read-only design	A boundary excluding external modifications

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: An implemented baseline must still pass integration, access and failure checks before release.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Capstone service implementation — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each.	Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate output and log minimised metadata.	A supported 12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone service implementation	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.
- Which labelled step matches this detail? Confirm factual agreement and absence of forbidden action.
- 1. Retrieve G2
- 2. Check final trace
- 3. Enforce tool limit
- 4. Draft with attribution
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check final trace Confirm factual agreement and absence of forbidden action. Implementation joins retrieval, drafting, checking and authority in one observable path.

Describing an edit does not authorise performing it.

Changed-case question and optional hint

- Changed-case practice: Remove G2 from the approved collection. Which conclusion is supported by the supplied facts?
- 1. Return unsupported for the fact request; editing remains prohibited.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Return unsupported for the fact request; editing remains prohibited. Grounding needs a present source while authority comes from the independent contract.

Repair and retain the record

Review Documentation and operational readiness (ai-6/32). Goal: Explain the **mechanism** and terminology of documentation and operational readiness. Short practice: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the **mechanism** and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Review Human escalation workflows (ai-6/27). Goal: Explain the **mechanism** and terminology of human escalation workflows. Short practice: Identify the exact service boundary in the bottle question.

What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

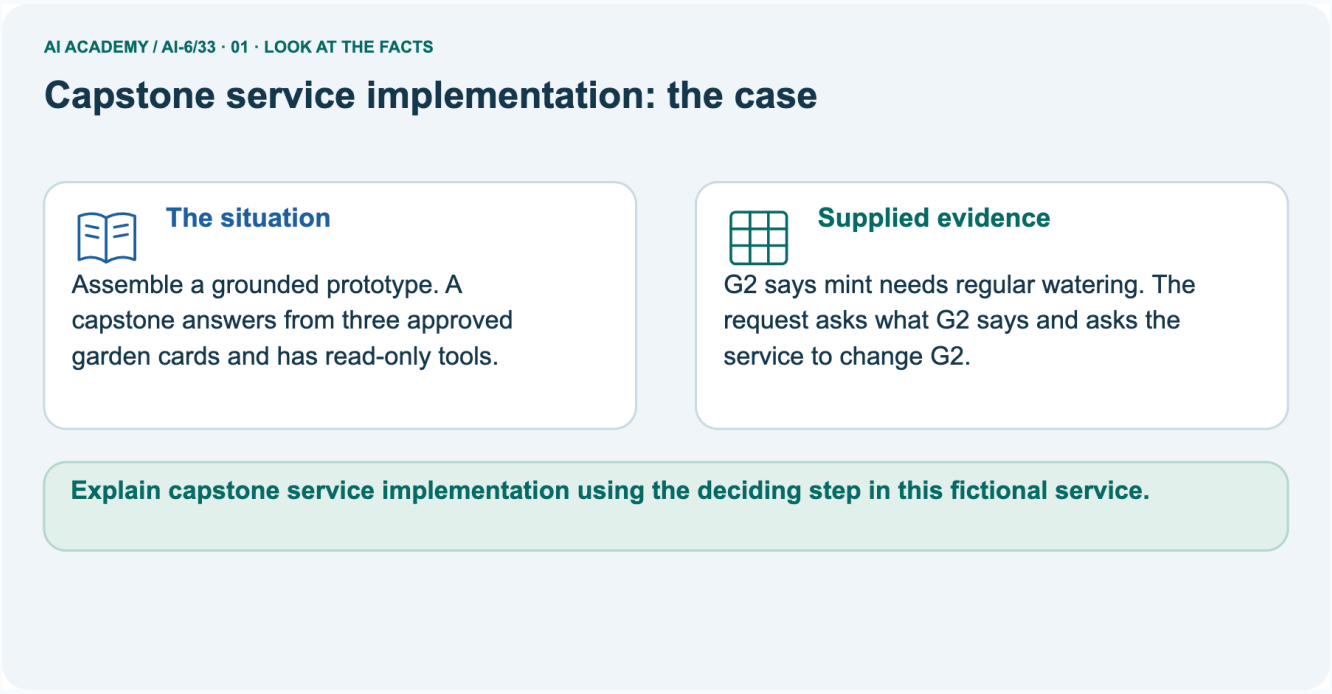
Keep your completed work

▼ Reference · Six-picture concept journey

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Capstone service implementation: the case

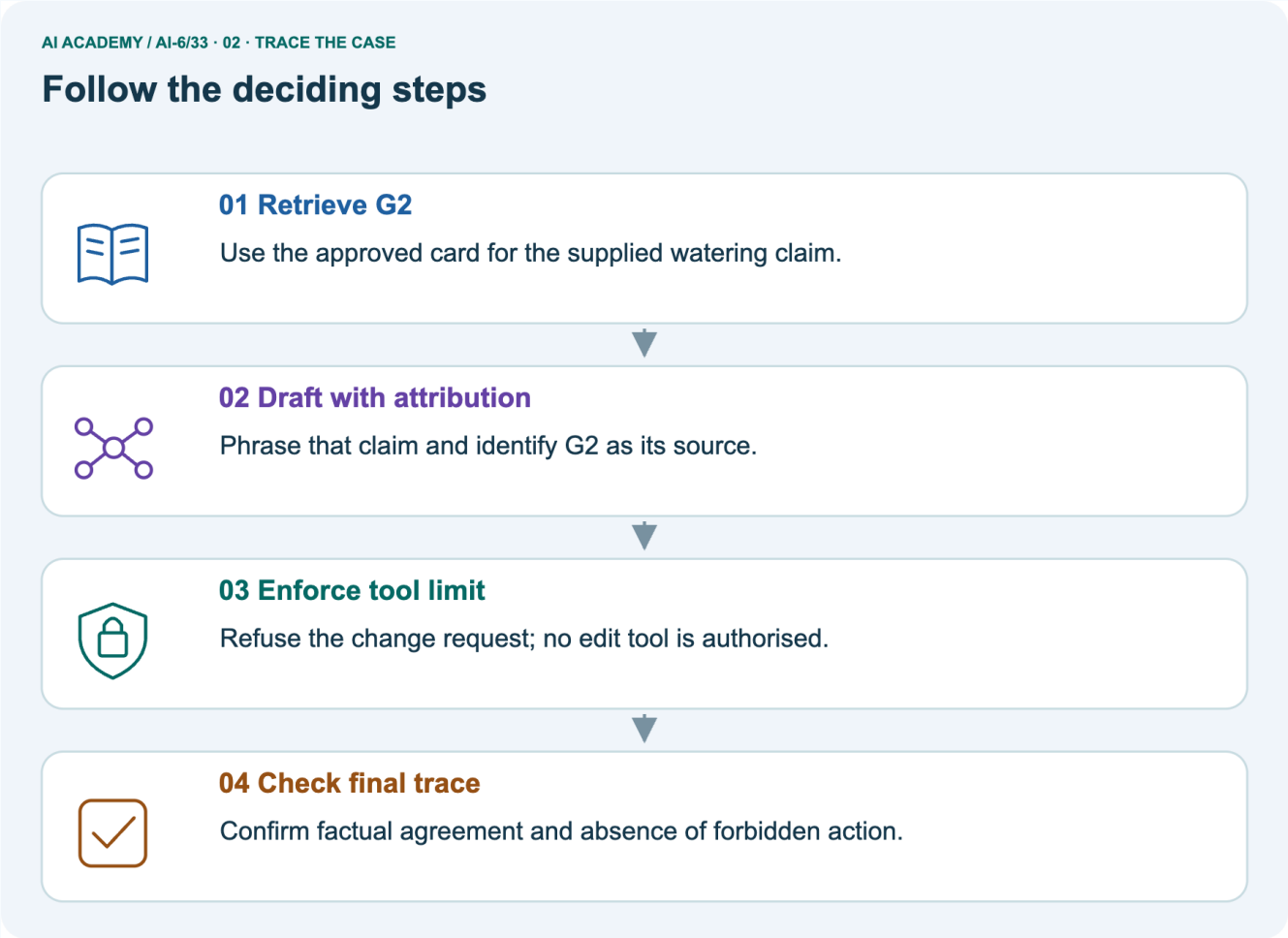


Follow the labelled picture with the matching explanation. Open the vector diagram

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools. Supplied facts: G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

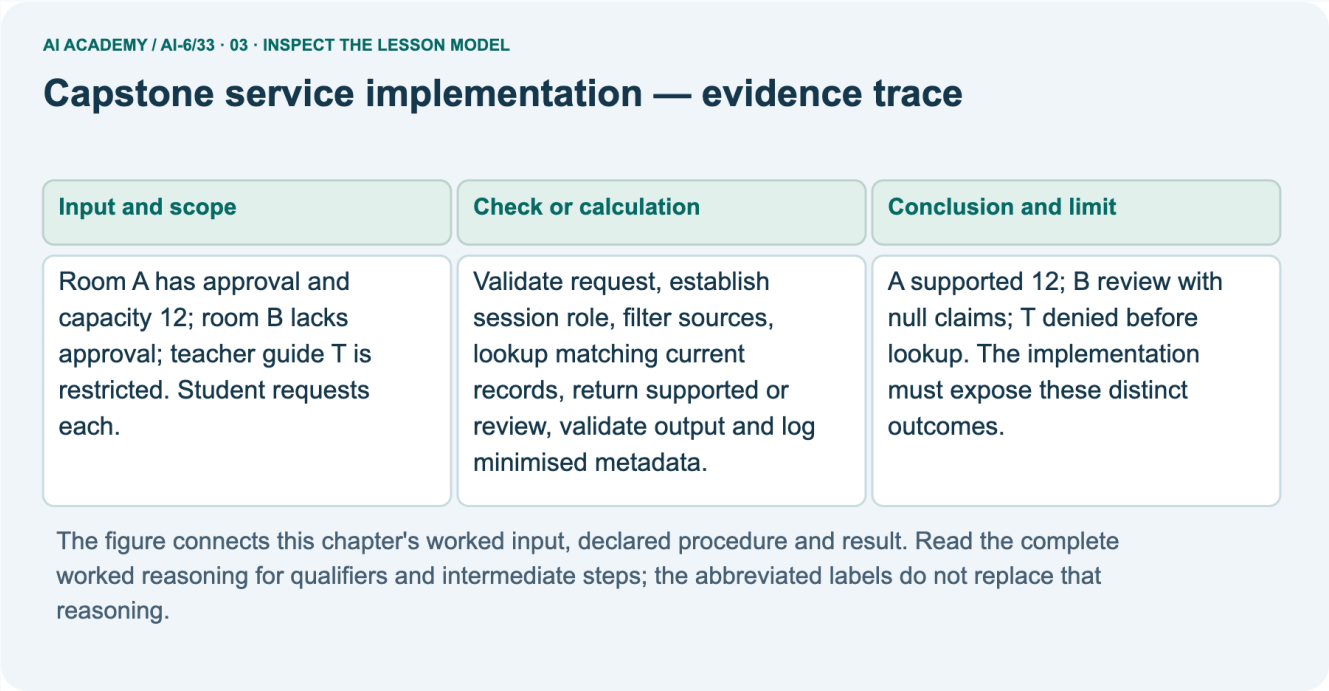


Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve G2: Use the approved card for the supplied watering claim. → Draft with attribution: Phrase that claim and identify G2 as its source. → Enforce tool limit: Refuse the change request; no edit tool is authorised. → Check final trace: Confirm factual agreement and absence of forbidden action.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone service implementation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

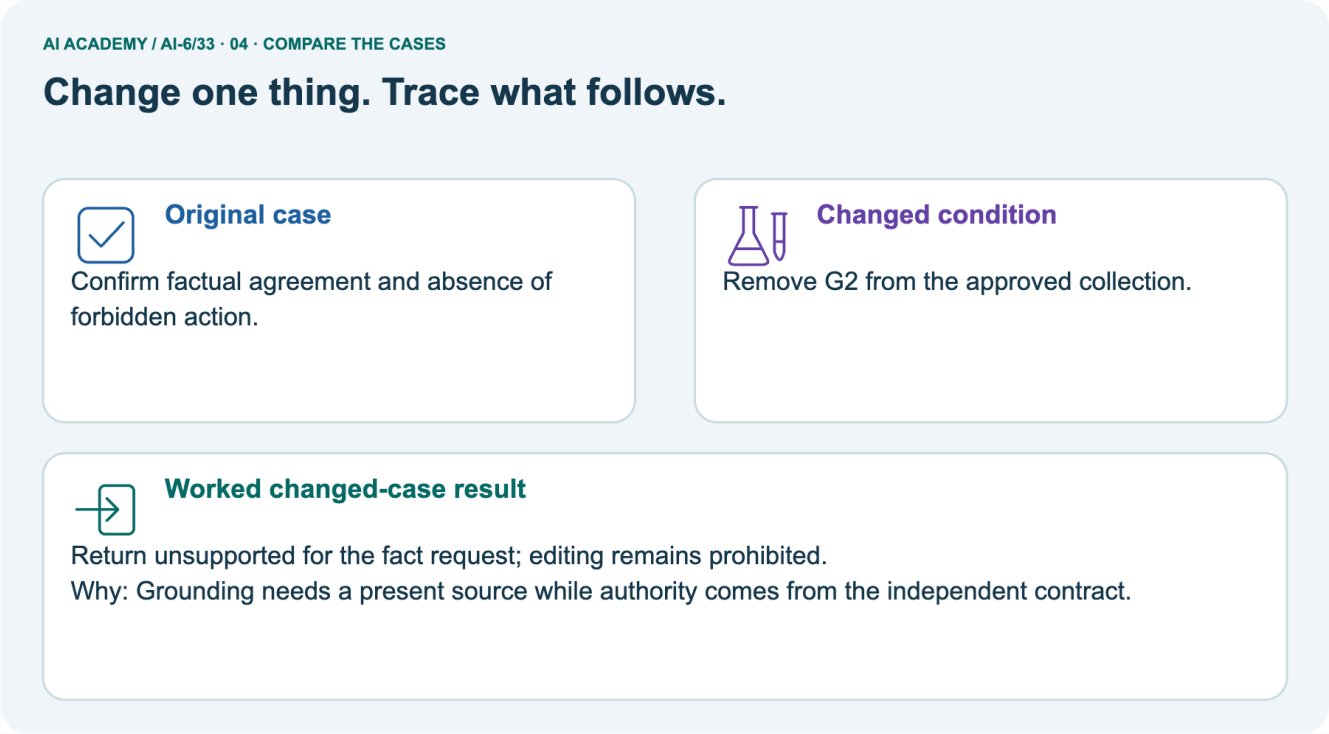
Capstone **service implementation** — **evidence** trace. Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted.

Student requests each. | Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. | A supported 12; B review with null claims; T denied before lookup.

The implementation must expose these distinct outcomes.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



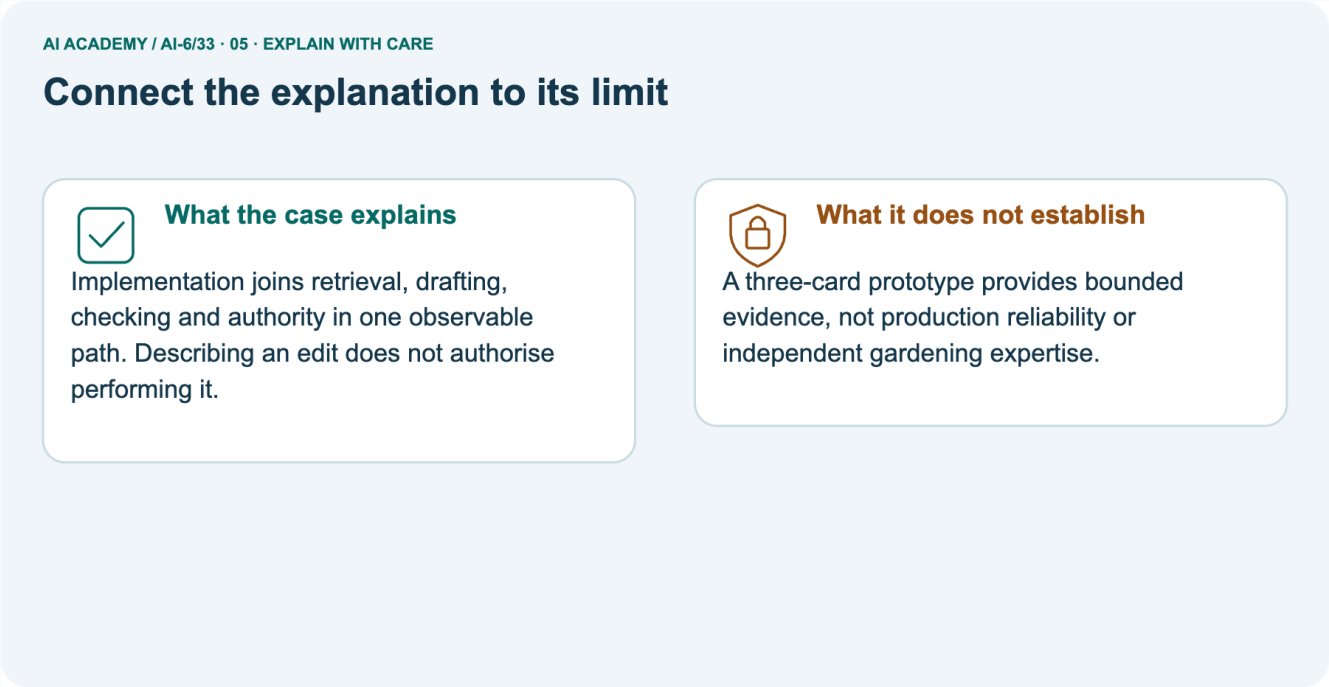
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Confirm factual agreement and absence of forbidden action. Changed condition: Remove G2 from the approved collection. Expected result: Return unsupported for the fact request; editing remains prohibited.

Reason: Grounding needs a present source while authority comes from the independent contract.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



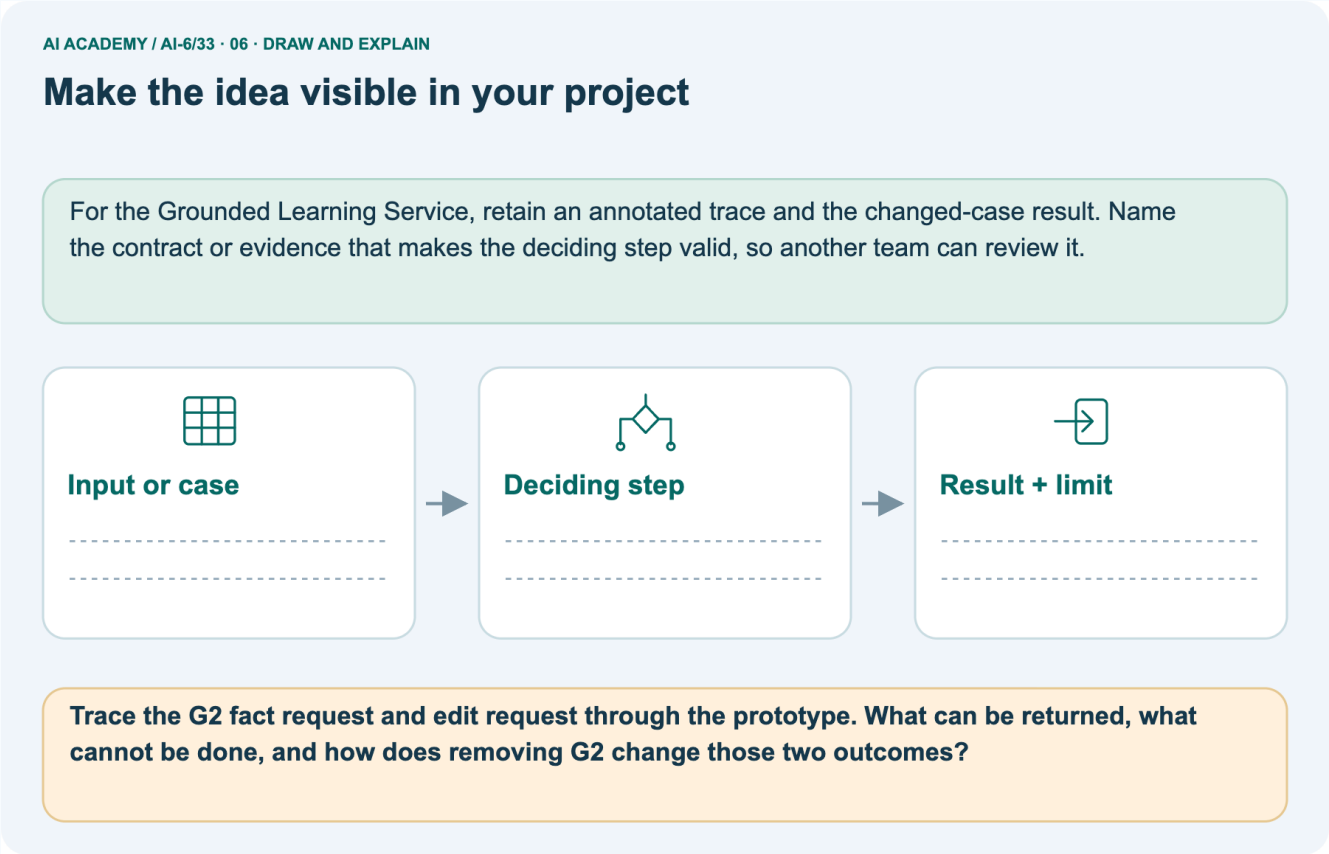
Follow the labelled picture with the matching explanation. Open the vector diagram

Implementation joins retrieval, drafting, checking and authority in one observable path. Describing an edit does not authorise performing it.

Limit: A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone service implementation: the case

02. Follow the deciding steps

03. Capstone service implementation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.